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A late Archaean radiating dyke swarm as possible clue to the origin of the Bushveld Complex

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The Bushveld Complex in South Africa represents the world's largest intrusion of magnesium- and iron-rich magmas. The Bushveld magmas were emplaced beneath the Transvaal basin[1] ~2.06 billion years ago[2,3], but their origin remains elusive. The magmas may have formed in response to an upwelling mantle plume[4], ancient subduction[5] or melting triggered by a meteorite impact[6]. Here we use U–Pb dating of baddeleyite crystals to date a series of mafic magmatic dykes located east of the Transvaal basin. We find that these dykes formed between 2.70 and 2.66 billion years ago, roughly 600 million years before the Bushveld magmas were emplaced. Collectively, the geometry of the dykes forms a radiating swarm converging towards a focal point in the eastern part of the Bushveld Complex. Such radiating swarms typically record the impact of a mantle plume head that injected large volumes of magma into the crust and at the base of the lithosphere. We propose that subsequent cooling and metamorphism of these mantle-plume-derived rocks caused them to increase in density and sink, triggering subsidence of the Transvaal basin. The dense rocks may later have sunk away into the mantle, with the delamination causing the inflow of hot mantle that initiated production of the voluminous Bushveld magmas about 600 million years after the mantle plume impact.

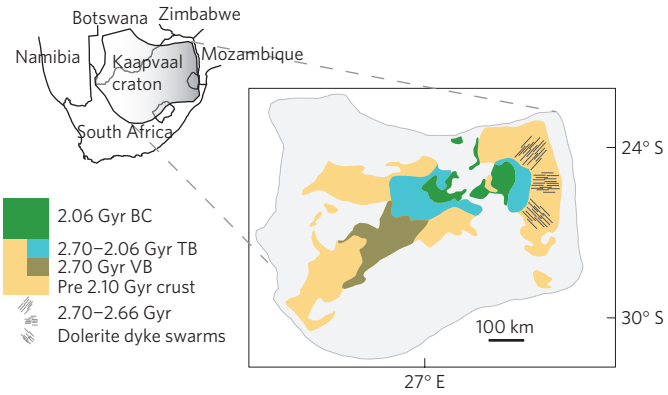


Figure 1 | Geological overview of the Kaapvaal craton. Generalized geological map of the Kaapvaal craton showing the present-day outlines of the ~2.70 Gyr Ventersdorp basin (VB), the ~2.70–2.06 Gyr Transvaal basin (TB), the 2.06 Gyr Bushveld Complex (BC) and the position of 2.70–2.66 Gyr NE-, E- and SE-trending dolerite dyke swarms in the Archaean basement east of the complex. The surrounding grey area is composed of younger lithological units.

The Kaapvaal craton in southern Africa is one of the oldest and best preserved cratons worldwide[7]. One of its most prominent geological features is the Bushveld Complex, which includes the largest known mafic layered intrusion in the world, the 7–9 km thick Rustenburg layered suite (RLS; ref. 1). The layered suite covers an area of ~65,000 km^2 and the preserved volume of mafic rocks in its western and eastern limbs is estimated to be up to 600,000 km^3, representing part of an original magma budget suggested to have been close to one million km^3 on the basis of the large chromium content in the Lower and Critical zones[8]. The RLS is also a major ore-bearing province for platinum-group elements (roughly 75% of the world's platinum resources), chromium, vanadium and nickel[9]. Most researchers postulate crustal contamination of up to ~40% to account for the geochemical and isotopic characteristics of the RLS intrusion (ref. 10 and references therein), although mixing of depleted asthenospheric mantle magma with sulphide components in the subcontinental lithospheric mantle (SCLM) has been suggested as an alternative mechanism for explaining radiogenic $^{187}Os/^{188}Os$ signatures in Bushveld rocks[11]. P-wave velocities below the Bushveld Complex on the order of 0.5–1.0% lower compared with surrounding SCLM (ref. 12), combined with reset Re–Os systematics in mantle nodules at ~2.05 billion years ago (Gyr; ref. 13), have been interpreted to reflect substantial chemical modification of the mantle during Bushveld magmatism. Radiometric dating, palaeomagnetic investigations and thermodynamic modelling collectively suggest that Bushveld Complex was emplaced and crystallized within 1–5 million years[2,3,8,14].

Until now, three general models for the origin of the Bushveld Complex have been proposed. As in many large igneous provinces (LIPs), a mantle plume origin has been considered[4] in which komatiitic to picritic primary melts[15] were emplaced into both the Archaean basement and the sediments of the Transvaal supergroup. However, previously hypothesized feeder dyke swarms[16] have been demonstrated to be older and, unlike most plume related LIPs, no radiating 2.06 Gyr dykes have yet been discovered in the Archaean basement around the Bushveld Complex. Furthermore, it has been questioned whether magma production corresponding in volume to the size of the Bushveld Complex could be achieved by the heat from a mantle plume at such great depths without significant lithospheric extension[17]. Structural investigations of Bushveld Complex and coeval intrusions have led some workers to argue that Bushveld magmas were derived in a back-arc setting related to tectonic activity in the Limpopo and Magondi orogenic belts[5]. Support for this comes from high Cl/F ratios and ^{37}Cl

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Table 1 | U–Pb baddeleyite analytical results.

Fraction (*n*)	Th/U	Pb*/Pbc	Atomic ratios $^{206}Pb/^{204}Pb$	$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	Corr. coeff.	Age (Myr) $^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	Discordance %
BCD5-22, NE dyke, Lat./Long. −23.912039/30.982789										
a (4)	0.071	10.0	760	11.99673±0.20538	0.494147±0.007932	0.95688	2588.6±34.2	2604.2±16.0	2616.3±8.3	1.3
b (5)	0.068	12.2	825	12.69884±0.10763	0.509808±0.003668	0.91036	2655.8±15.7	2657.6±8.0	2658.9±5.9	0.1
c (4)	0.053	4.2	329	12.29248±0.21236	0.499088±0.006555	0.85692	2609.9±28.2	2627.0±16.2	2640.2±15.1	1.4
d (4)	0.155	14.5	1023	12.75804±0.12665	0.512116±0.004739	0.95744	2665.7±20.2	2662.0±9.3	2659.1±4.8	−0.3
e (5)	0.044	4.8	416	12.44938±0.25934	0.505013±0.009554	0.94122	2635.3±40.9	2638.9±19.5	2641.7±11.8	0.3
BCD5-27, NE dyke, Lat./Long. −23.664522/30.740119										
a (4)	0.103	5.0	391	10.62282±0.15241	0.456924±0.004785	0.83706	2426.0±21.2	2490.7±13.3	2543.9±13.4	5.6
b (8)	0.102	4.2	341	11.47167±0.14673	0.477496±0.003136	0.75467	2516.4±13.7	2562.3±12.0	2598.8±15.0	3.8
c (7)	0.124	12.5	859	11.63620±0.07938	0.482735±0.002533	0.86371	2539.2±11.0	2575.6±6.4	2604.4±5.8	3.0
d (8)	0.095	15.5	990	11.18946±0.07517	0.471209±0.002576	0.88742	2488.9±11.3	2539.0±6.3	2579.4±5.2	4.2
e (7)	0.113	16.8	1126	11.67043±0.07305	0.481936±0.002544	0.89616	2535.7±11.1	2578.3±5.9	2612.0±4.7	3.5
f (7)	0.101	19.8	1338	10.99091±0.07739	0.468107±0.002974	0.93516	2475.3±13.1	2522.4±6.5	2560.5±4.2	4.0
BCD5-23, NE dyke, Lat./Long. −23.923181/30.891761										
a (1)	0.021	13.5	956	12.57309±0.09501	0.503869±0.003277	0.91479	2630.4±14.0	2648.2±7.1	2661.8±5.1	1.4
b (3)	0.028	9.4	665	11.61704±0.08726	0.475847±0.002479	0.80702	2509.2±10.8	2574.0±7.0	2625.5±7.5	5.3
BCD3-08, NE dyke, Lat./Long. −26.567481/30.830489										
a (3)	0.053	31.9	2012	13.05562±0.04234	0.513535±0.001240	0.88671	2671.7±5.3	2683.7±3.1	2692.7±2.6	1.0
b (5)	0.134	6.2	463	13.07862±0.10612	0.514420±0.001535	0.79688	2675.5±6.5	2685.3±7.7	2692.8±10.0	0.8
c (6)	0.137	46.2	2954	12.38550±0.03386	0.487283±0.001028	0.89303	2558.9±4.5	2634.1±2.6	2692.4±2.1	6.0
d (1)	0.162	9.4	663	13.09949±0.16071	0.514305±0.005666	0.93703	2675.0±24.1	2686.8±11.6	2695.8±7.1	0.9
e (2)	0.121	21.6	1333	13.06114±0.07936	0.514228±0.002759	0.93020	2674.7±11.7	2684.1±5.7	2691.2±3.7	0.8
BCD1-04, NE dyke, Lat./Long. −25.716461/30.950300										
a (2)	0.298	133.0	7273	13.01248±0.03577	0.512362±0.001238	0.93179	2666.7±5.3	2680.6±2.6	2691.0±1.7	1.1
b (5)	0.186	11.2	739	12.83325±0.07790	0.505956±0.001874	0.80184	2639.3±8.0	2667.5±5.7	2688.9±6.3	2.2
c (2)	0.095	46.0	2699	12.42045±0.04767	0.494443±0.001672	0.93760	2589.9±7.2	2636.7±3.6	2672.9±2.2	3.8
d (2)	0.188	73.0	4041	12.63079±0.04449	0.500803±0.001590	0.94808	2617.2±6.8	2652.5±3.3	2679.5±1.9	2.8

Abbreviations are: (*n*)—number of crystals in each fraction, Lat. and Long.—latitude and longitude of sample localities. Pb* is radiogenic lead and Pbc is the total measured common lead with assumed isotopic composition of the laboratory blank ($^{206}Pb/^{204}Pb$—18.221; $^{207}Pb/^{204}Pb$—15.612; $^{208}Pb/^{204}Pb$—39.360). Atomic Pb/U ratios are corrected for spike, fractionation, blank and initial common lead. Absolute errors of atomic ratios and errors of age calculations are reported at the 95% confidence level. The measured $^{206}Pb/^{204}Pb$ is corrected for spike and fractionation.

values recorded in the Main and Critical zones[18], a boninitic affinity for marginal rocks, and from calc-alkaline compositions of volcanics in the Transvaal basin[19]. However, a subduction setting is at odds with high $\delta^{18}O$ values recorded in the RLS (ref. 10). We also note that the Bushveld Complex and coeval satellite intrusions differ in composition, style and temporal evolution from igneous batholith complexes emplaced over tens to hundreds of Myr along active continental margins. A meteorite impact model has also been suggested[6], whereby high $\delta^{18}O$ values could be explained by assimilation of upper crustal melts[10], but the lack of unambiguous evidence of shocked quartz in the country rocks[20] and other features typical for bolide impacts renders this model unlikely.

Three distinct dolerite dyke swarms radiate northeast (NE), east (E) and southeast (SE) through the Archaean crust east of the Bushveld Complex and converge towards an apparent locus situated in the eastern lobe of the complex (Fig. 1). Recent U–Pb (ID-TIMS; isotope-dilution—thermal ionization mass spectrometry; see Supplementary Information for details on the methods used) dating studies have defined emplacement ages of about 2.96 Gyr for dykes of the SE-trending swarm, 2.69–2.66 Gyr for the E-trending swarm (the Rykoppies swarm)[3] and 1,850 ± 30 Myr for the NE-trending swarm (Johan Olsson, personal communication). Here, we report further 2.70–2.66 Gyr emplacement ages for three NE- and two SE-trending dykes based on analyses of minute grains (typically <50 μm in long dimension) of magmatic baddeleyite (ZrO_2) (see Table 1 for U–Pb data). Together with published data from five E-trending dykes of the Rykoppies swarm[3], sets of 2.70–2.66 Gyr dykes now define a major radiating swarm with a broad fanning arc of about 90°. Data for NE-trending dyke samples near the town of Phalaborwa (BCD5-22 and BCD5-27) yield ages of 2,659 ± 13 Myr and 2,701 ± 11 Myr, respectively, and are clearly contemporaneous with the better-defined E–W Rykoppies subswarm (Fig. 2). Sparse baddeleyite grains recovered from a NE-dyke in the same area (BCD5-23) yield an upper intercept age of 2,674 ± 11 Myr, confirming that this dyke also belongs to the same generation (Fig. 2). Free regression of data for BCD5-22 and BCD5-27 yield lower intercepts close to about 1.9 Gyr, similar to the emplacement age of younger NE-trending dykes in the same region. On the assumption that lead loss in baddeleyite (or partial replacement of baddeleyite to zircon) was temporally linked to the intrusion of these dykes, we have anchored lower intercepts at 1,850 ± 30 Myr in the regression of these samples. In the southern part of the study area, the ages of SE-trending dykes in the vicinity of the towns of Amsterdam and Barberton are dated at 2,692 ± 1.4 Myr (BCD3-08) and 2,698 ± 3.6 Myr (BCD1-04), respectively (Fig. 2).

The spatial coincidence between the 2.70 Gyr focal point of a radiating dyke swarm, the thickest parts of the Transvaal basin, and the eastern lobe of the 2.06 Gyr Bushveld Complex lead to the following tectonomagmatic model, outlined in Fig. 3. We propose that a 2.70 Gyr mantle plume induced voluminous mantle melting which first erupted as the Klipriviersberg group (komatiitic) basalts and subsequently injected magmas into a large reservoir at depth

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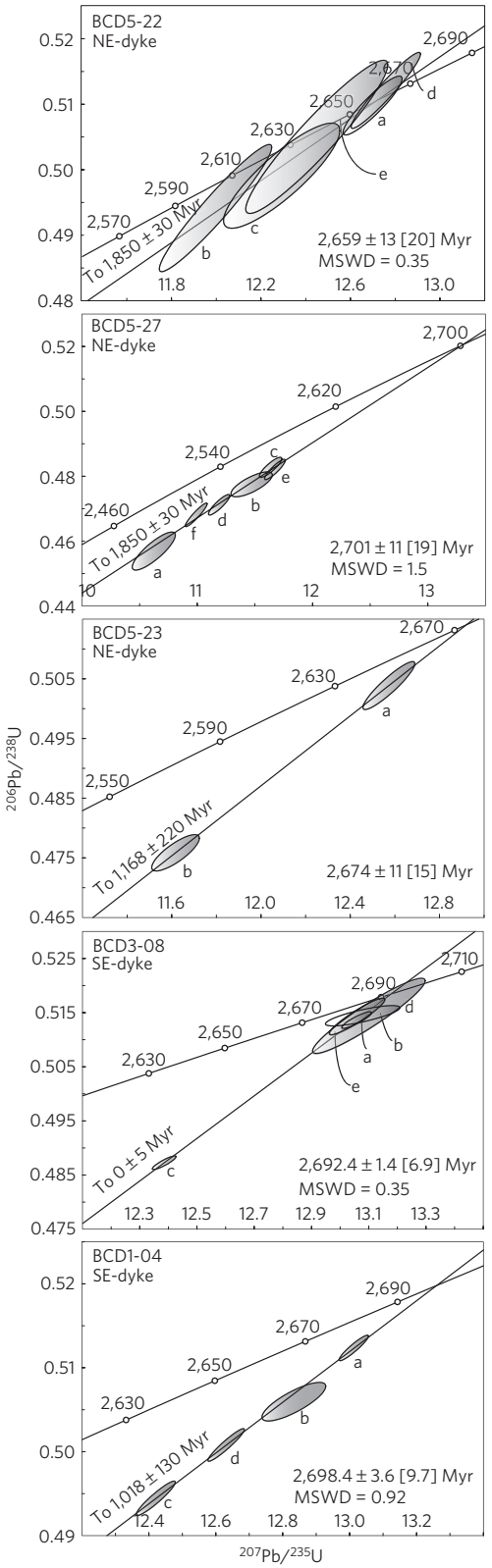


Figure 2 | U–Pb geochronology of mafic dykes. Concordia diagrams with $^{206}Pb/^{238}U$ versus $^{207}Pb/^{235}U$ isotope ratios with upper intercept ages in the lower right corner and lower intercept ages above regression lines. Age errors that include decay constant uncertainties are shown within square brackets. Collectively with previously published age data[3] the results define an age span of 2.70–2.66 Gyr. Error ellipses and age calculations are shown at the 95% confidence level. MSWD represents the mean square of weighted deviates.

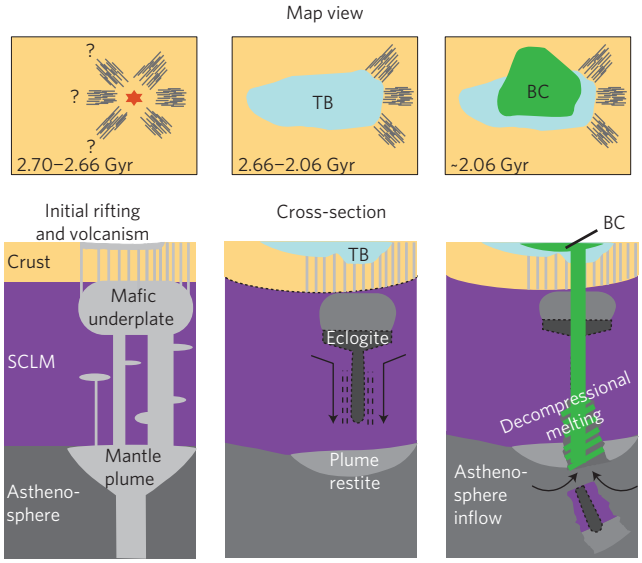


Figure 3 | The origin of the Bushveld Complex. Schematic map views (top) and cross-sections (bottom) illustrating the ~2.70 Gyr mantle plume impact, the subsequent evolution of the Transvaal basin (TB) and the delamination-triggered origin of 2.06 Gyr Bushveld Complex (BC) magmas. Question marks in the map view label undiscovered westerly swarm segments, yet SW–NE trending dykes in the southwestern corner have been inferred as feeders to the Klipriviersberg group flood basalts[25]. The cross-sections are not to scale and are vertically exaggerated to show the details. See main text for a detailed description of the model.

(collectively viewed as mafic underplate in Fig. 3) from which dolerite dykes were fed. Crustal doming was accompanied by the emplacement of these magmas in at least three directions that define a radiating dyke pattern—one of the principal criteria of LIPs related to mantle plumes[21]. When extrapolated westwards, the radiating dolerite dykes converge towards an apparent focus coinciding with the eastern lobe of the Bushveld Complex, and fix the locus of Neoarchaean mafic magmatism in this singular region. Cooling of injected magmas throughout the lithosphere led to negative buoyancy-driven downwelling and concomitant eclogitization in the SCLM, which may have facilitated the formation of the Transvaal basin and its infilling of sediments from about 2.70 to 2.06 Gyr (ref. 22). At 2.06 Gyr, tectonic activity along the northern and northwestern margins associated with the Limpopo and the Magondi orogens (not shown in Fig. 3) triggered decoupling and foundering of the dense mantle root. Decompressional melting of the asthenospheric mantle, ascending up into the delaminated void, and any SCLM and less refractory mafic components that were sufficiently heated by the upwelling asthenosphere produced large magma volumes. These primary magmas would have ascended rapidly up through the SCLM and lower crust along pre-existing zones of weakness (for example the Murchinson–Thabazimbi lineament[5]) before being emplaced into the Transvaal sedimentary units as the large lopolithic Bushveld Complex.

The dolerite dykes of the radiating swarm seem to be both temporally and petrogenetically related to lavas from the uppermost part of the Ventersdorp supergroup (Allanridge formation)[23], as well as share a common palaeomagnetic pole[24]. Voluminous flood basalts from the lower part of the Ventersdorp supergroup (Klipriviersberg group), with basal ~2,714 Myr komatiitic compositions, have themselves been attributed to a plume impact[4], primarily fed through a dense swarm of SW–NE trending dykes[25]. It is likely that the same plume source gave rise to both the ~2,714 Myr flood basalt event and the 2.70–2.66 Gyr radiating

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dyke swarm, all of which converged onto what was to become the eastern basin/lobe of the Transvaal/Bushveld Complex. Early rifting of the Transvaal basin during a period with otherwise scant evidence of active plate processes[22] was controlled by brittle deformation of the upper crust, in response to viscous drag at lower levels by a progressively denser, sinking SCLM—first during cooling (the fluvial sedimentation of the Black Reef formation and the marine deposits of the Chuniespoort group) and then eclogitization (Pretoria group) of plume-derived mafic intrusions. This scenario is in agreement with seismic investigations inferring a crustal thickness approximately 5 km greater beneath the Bushveld Complex than the surrounding region[26], but also rationalizes the apparent greater subsidence of the eastern Transvaal basin relative to its western half during deposition of Pretoria group sediments and volcanics[22]. Such an evolutionary model for the Transvaal basin is similar to the conclusions of thermal modelling of the large Phanerozoic intracratonic basins of North America (for example the Hudson Bay, Williston, Michigan and Illinois basins), where plume-derived mafic intrusions also controlled basin subsidence[27].

Granulite-facies metamorphism in the Magondi belt at the northwestern flank of the Zimbabwe craton has been bracketed between 2,027 and 1,960 Myr (ref. 28), during which time the northern margin of the Kaapvaal craton (Central zone of the Limpopo belt) was subjected to high-grade metamorphism and development of dextral transpressional shear zones[29]. The combined stresses of these tectonic events represent a possible trigger for foundering of a dense lithospheric root. Lower P-wave velocities[12] and resetting of isotopic systems[13] would in our model reflect the consequent and coincident inflowing of asthenospheric mantle and subsequent chemical modification of the SCLM at the time of detachment of the eclogitized dense root at 2.06 Gyr. Cr-rich cumulates within any incorporated mafic components in the remelted SCLM offer an alternative explanation for the enigmatic large chromium budget of the Bushveld Complex magmas. Furthermore, magmas composed of a mixture of asthenospheric and SCLM material is in accordance with sulphide Re–Os and silicate Sm–Nd and Rb–Sr isotope compositions from Bushveld rocks[11]. A similar lower SCLM delamination model has been suggested for the ~250 Myr Siberian flood basalts[30], where an estimated one million km^3 of magma emplaced within 1 Myr is comparable to the proposed volumes and maximum magma production rate during the formation of the Bushveld Complex (ref. 8).

The existence of Archaean and Proterozoic diamonds from kimberlites sampling depths in excess of 200 km in the Bushveld Complex region[11] would seem to contradict the suggestion of removal of pre-Bushveld SCLM. However, with the exception of the ~ 100 Myr Palmietgat kimberlite[31], most diamond-bearing kimberlites are located along the marginal zone to the Bushveld Complex, such as the ~ 1.2 Gyr Premier kimberlite on the southern flank of the Bushveld Complex. It is possible that thicker SCLM was preserved at the margins of the complex and/or a more 'chaotic' delamination maintained some 'pockets' of diamondiferous SCLM within its central parts, hosting diamonds for later kimberlite eruptions. With this caveat, our model remains attractive in explaining the distinct spatial connection between the Bushveld Complex, the Transvaal basin and the centre of a 600 Myr older, late Archaean Ventersdorp LIP.

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References

1. Harney, D. M. W. & von Gruenewaldt, G. Ore-forming processes in the upper part of the Bushveld complex, South-Africa. *J. Afr. Earth Sci.* **20,** 77–89 (1995).
2. Scoates, J. S. & Friedman, R. M. Precise age of the platiniferous Merensky Reef, Bushveld Complex, South Africa, by the U–Pb zircon chemical abrasion ID-TIMS technique. *Econ. Geol.* **103,** 465–471 (2008).
3. Olsson, J. R., Söderlund, U., Klausen, M. B. & Ernst, R. E. U–Pb baddeleyite ages linking major Archean dyke swarms to volcanic-rift forming events in the Kaapvaal craton (South Africa), and a precise age for the Bushveld Complex. *Precambr. Res.* **183,** 490–500 (2010).
4. Hatton, C. J. Mantle plume origin for the Bushveld and Ventersdorp magmatic provinces. *J. Afr. Earth. Sci.* **21 (4),** 571–577 (1995).
5. Clarke, B., Uken, R. & Reinhardt, J. Structural and compositional constraints on the emplacement of the Bushveld Complex, South Africa. *Lithos* **111,** 21–36 (2009).
6. Rhodes, R. C. New evidence for impact origin of the Bushveld Complex, South Africa. *Geology* **3,** 549–554 (1975).
7. de Wit, M. J. *et al.* Formation of an Archaean continent. *Nature* **357,** 553–562 (1992).
8. Cawthorn, R. G. & Walraven, F. Emplacement and crystallization time for the Bushveld Complex. *J. Petrol.* **39,** 1669–1687 (1998).
9. Lee, C. A. in *Layered Intrusions* (ed. Cawthorn, R .G) 103–145 (Elsevier, 1996).
10. Harris, C., Pronost, J. J. M., Ashwal, L. D. & Cawthorn, R. G. Oxygen and hydrogen isotope stratigraphy of the Rustenburg layered suite, Bushveld Complex: Constraints on crustal contamination. *J. Petrol.* **46,** 579–601 (2005).
11. Richardson, S. H. & Shirey, S. B. Continental mantle signature of Bushveld magmas and coeval diamonds. *Nature* **453,** 910–913 (2008).
12. James, D. E., Fouch, M. J., VanDecar, J. C., van der Lee, S. & Kaapvaal Seismic Group, Tectospheric structure beneath southern Africa. *Geophys. Res. Lett.* **28,** 2485–2488 (2001).
13. Carlson, R. W. *et al.* Continental growth, preservation and modification in southern Africa. *GSA Today* **10,** 1–7 (2001).
14. Letts, S., Torsvik, T. H., Webb, S. J. & Ashwal, L. D. Palaeomagnetism of the 2054 Ma Bushveld Complex (South Africa): Implications for emplacement and cooling. *Geophys. J. Int.* **179,** 850–872 (2009).
15. Barnes, S. J. Are Bushveld U-type parent magmas boninites or contaminated komatiites? *Contrib. Mineral. Petrol.* **101,** 447–457 (1989).
16. Uken, R. & Watkeys, M. K. An interpretation of mafic dyke swarms and their relationship with major mafic magmatic events on the Kaapvaal Craton and Limpopo Belt. *South Afr. J. Geol.* **100,** 341–348 (1997).
17. Elkins-Tanton, L. T. Continental magmatism caused by lithospheric delamination. *Spec. Pap. Geol. Soc. Am.* **388,** 449–461 (2005).
18. Willmore, C. C., Boudreau, A. E., Spivack, A. & Kruger, F. J. Halogens of Bushveld Complex, South Africa: δ^{37} Cl and Cl/F evidence for hydration melting of the source region in a back-arc setting. *Chem. Geol.* **182,** 503–511 (2002).
19. Harmer, R. E. & von Gruenewaldt, G. A review of magmatism associated with the Transvaal Basin — implications for its tectonic setting. *South Afr. J. Geol.* **94,** 104–122 (1991).
20. Buchanan, P. C. & Reimold, W. U. Studies of the Rooiberg group, Bushveld Complex, South Africa: No evidence for an impact origin. *Earth Planet. Sci. Lett.* **155,** 149–165 (1998).
21. Ernst, R. E. & Buchan, K. L. in *Mantle Plumes: Their Identification Through Time* Vol. 352 (eds Ernst, R. E. & Buchan, K. L.) 247–265 (Geological Society of America Special Paper, 2001).
22. Eriksson, P. G., Altermann, W., Catuneanu, O., van der Merwe, R. & Bumby, A. J. Major influences on the evolution of the 2.67—2.1 Ga Transvaal basin, Kaapvaal craton. *Sedim. Geol.* **141–142,** 205–231 (2001).
23. Klausen, M. B. *et al.* Petrological discrimination among dated Precambrian dyke swarms, Eastern Kaapvaal craton (South Africa). *Precambr. Res.* **183,** 501–522 (2010).
24. Lubnina, N., Ernst, R. E., Klausen, M. B. & Söderlund, U. Paleomagnetic study of Neoarchean-Paleoproterozoic dykes in the Kaapvaal Craton. *Precambr. Res.* **183,** 523–552 (2010).
25. McCarthy, T. S, McCallum, K., Myers, R. E. & Linton, P. Stress states along the northern margin of the Witwatersrand Basin during Klipriviersberg Group volcanism. *South Afr. J. Geol.* **93,** 245–260 (1990).
26. Wright, C. Station corrections for the Kaapvaal seismic network: Statistical properties and relation to lithospheric structure. *Phys. Earth Planet. Inter.* **167,** 39–52 (2008).
27. Kaminski, E. & Jaupart, C. Lithosphere structure beneath the Phanerozoic intracratonic basins of North America. *Earth Planet. Sci. Lett.* **178,** 139–149 (2000).
28. Mapeo, R. B. M., Armstrong, R. A. & Kampunzu, A. B. SHRIMP U–Pb zircon geochronology of gneisses from the Gweta borehole, northeast Botswana: Implications for the Palaeoproterozoic Magondi Belt in southern Africa. *Geol. Mag.* **138,** 299–308 (2001).
29. Holzer, L., Frei, R., Barton, J. M. & Kramers, J. D. Unravelling the record of successive high-grade events in the Central Zone of the Limpopo Belt using single phase dating of metamorphic minerals. *Precambr. Res.* **87,** 87–115 (1999).

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30. Elkins-Tanton, L. T. & Hager, B. H. Melt intrusion as trigger for lithospheric foundering and the eruption of the Siberian flood basalts. *Geophys. Res. Lett.* **27** (23), 3937–3940 (2000).
31. Webb, S. J., Ashwal, L. D. & Cawthorn, R. G. Continuity between eastern and western Bushveld Complex, South Africa, confirmed by xenoliths from kimberlites. *Contrib. Mineral. Petrol.* **162,** 101–107 (2011).

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Author contributions

M.B.K. and U.S. initiated the study and sampled the targeted dolerite dykes. J.R.O. processed the dyke samples and extracted the baddeleyite separates. M.A.H. and J.R.O. performed the U–Pb ID-TIMS analyses at the Jack Satterly Geochronology Laboratory, University of Toronto. G.R.H. provided important re-interpretations of published seismic data. J.R.O. and U.S. wrote the paper. All authors discussed the results and commented on the manuscript.

Additional information

The authors declare no competing financial interests. Supplementary information accompanies this paper on www.nature.com/naturegeoscience. Reprints and permissions information is available online at http://www.nature.com/reprints. Correspondence and requests for materials should be addressed to J.R.O.

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